

45GP004

POLYETHYLENE RESIN FOR PIPE EXTRUSION

Relene 45GP004 (PE:80) is a natural coloured grade for pipe extrusion.

It possesses bimodal molecular weight distribution, which improves processability. This grade meets the requirements of material as per clause 5.2 of IS 4984:2016 for 'Polyethylene pipes for Water Supply - Specification'. When incorporated with adequate quantity of carbon black masterbatch as specified in clause 5.3 of IS: 4984: 2016, pipes made with this grade confirms to all specifications as per IS 4984- 2016.

Pipe made from Relene 45GP004 grade passes the Internal Pressure Creep Rupture Test when tested with 4.5 MPa induced stress at 80°C for 165 hrs and with 4.9 MPa induced stress at 80°C for 48 hrs as per clause 8.1 of IS 4984:2016, thereby meeting the requirement of PE:80 grade.

This grade conforms to ISO 4427 specifications shown below and has been certified for MRS > 8 MPa as per ISO 9080.

BIS Designation Code: IS 7328-3B-PB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Hydrostatic Pressure Test	ISO 1167	Hrs	
10 MPa @ 20°C			>100
4.5 MPa @ 80°C			>165
4.0 MPa @ 80°C			>1000
Melt Flow Index (190°C/ 5.0 kg)	ASTM D1238	g/10 min	0.45
Density (23°C)	ASTM D792	g/cm ³	0.948
Thermal Stability (O.I.T.) 200°C	ASTM D3895	Min	>30
Tensile Stress at Yield	ASTM D638	MPa	27
Elongation at Break	ASTM D638	%	>600
Flexural Modulus	ASTM D790	MPa	850
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR @ 10% Igepal Soln (F ₅₀)	ASTM D1693B	hrs	>1000

* Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Typical Processing Temperature: 180 – 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

- PE80 Pressure Pipes, Sheets.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

The level of antioxidant in 45GP004 is not more than 0.3% by mass of the resin.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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46GP003

POLYETHYLENE RESIN FOR PIPE EXTRUSION

Relene 46GP003 (PE:100) is a natural coloured grade for pipe extrusion.

It possesses bimodal molecular weight distribution which improves processability. This grade meets the requirements of material as per clause 5.2 of IS 4984:2016 for 'Polyethylene pipes for Water Supply - Specification'. When incorporated with adequate quantity of carbon black masterbatch as specified in clause 5.3 of IS: 4984: 2016, pipes made with this grade confirms to all specifications as per IS 4984- 2016.

Pipe made from Relene 46GP003 grade passes the Internal Pressure Creep Rupture Test when tested with 5.4 MPa induced stress at 80°C for 165 hrs and with 5.7 MPa induced stress at 80°C for 48 hrs as per clause 8.1 of IS 4984:2016, thereby meeting the requirement of PE:100 grade.

This grade conforms to ISO 4427 specifications shown below and has been certified for MRS > 10 MPa as per ISO 9080.

BIS Designation Code: IS 7328-3B-PB-ETA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Hydrostatic Pressure Test	ISO 1167	Hrs	
12.4 MPa @ 20°C			>100
5.4 MPa @ 80°C			>165
5.0 MPa @ 80°C			>1000
MRS	ISO 9080	MPa	>10.0
SCG	ISO 13479	Hrs	>500
Melt Flow Index (190°C/ 5.0 kg)	ASTM D1238	g/10 min	0.30
Density (23°C)	ASTM D792	g/cm ³	0.948
Thermal Stability (O.I.T.) 200°C	ASTM D3895	Min	>30
Tensile Stress at Yield	ASTM D638B	MPa	28
Elongation at Break	ASTM D638B	%	>600

Flexural Modulus	ASTM D790	MPa	850
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR @ 10% Igepal Soln (F ₅₀)	ASTM D1693B	hrs	>1000

* Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Typical Process Temperature: 180 – 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

PE100 Pressure Pipes, Sheets.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

The level of antioxidant in 46GP003 is not more than 0.3% by mass of the resin.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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46GP009

POLYETHYLENE RESIN

FOR PIPE EXTRUSION AND PESTICIDE BOTTLES

Relene 46GP009 is a natural coloured PE:63 grade for pipe extrusion. It possesses bimodal molecular weight distribution which improves processability and mechanical properties. This grade meets the requirements of material as per clause 5.2 of IS 4984:2016 for 'Polyethylene pipes for Water Supply - Specification'. When incorporated with adequate quantity of carbon black masterbatch as specified in clause 5.3 of IS: 4984: 2016, pipes made with this grade confirms to all specifications as per IS 4984-2016.

The pipe made from this grade meets the hydrostatic strength requirement of PE-63 class of pipe as per IS 4984:2016.

The grade has excellent ESCR properties and is suitable for pesticide bottles.

BIS Designation Code: IS 7328-3B-PB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Index (190°C/ 5.0 kg)	ASTM D1238	g/10 min	0.9
Density (23°C)	ASTM D792	g/cm ³	0.948
Tensile Stress at Yield	ASTM D638	MPa	24.0
Elongation at Yield	ASTM D638	%	11.0
Elongation at Break	ASTM D638	%	>600
ESCR @ 10% Igepal Soln (F ₅₀)	ASTM D1693B	hrs	>1000

* Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Typical Process Temperature: 180 – 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Pipes for drinking water, drip irrigation, sprinkler systems, industrial effluent etc. , Pesticide bottles, Sheets, DWC Pipes.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

The level of antioxidant in 46GP009 is not more than 0.3% by mass of the resin.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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46GP009UV

POLYETHYLENE RESIN FOR TELECOM DUCTS

Relene 46GP009UV is a natural HDPE extrusion grade & possesses bimodal molecular weight distribution which improves process ability and mechanical properties of pipes & ducts.

46GP009UV is UV stabilized grade & is suitable for producing Telecom ducts to be used as Underground Optical fiber cable & conduits and for other general-purpose application. The grade is designed to sustain outdoor exposure of 12 months. 46GP009UV conforms to TEC generic requirements No. TEC 72030:2025 & TEC 72020:2019 (TEC/GR/FA/DWC-034/02/AUG-19).

BIS Designation Code: IS 7328-3B-PB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Index (190°C/ 5.0 kg)	ASTM D1238	g/10 min	0.9
Density (23°C)	ASTM D792	g/cm ³	0.948
Thermal Stability (O.I.T.) 200°C	ASTM D3895	Min	>30
Tensile Stress at Yield	ASTM D638	MPa	24.0
Elongation at Yield	ASTM D638	%	11
Elongation at Break	ASTM D638	%	>600
Flexural Modulus	ASTM D790	MPa	750
Hardness	ASTM D2240	Shore D	63
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	72
ESCR @ 10% Igepal Soln (F50)	ASTM D1693B	Hrs	>1000

* Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Typical Process Temperature: 180 – 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Ducts for optical fiber cables, DWC Pipes, Sheets & General-purpose applications

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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52GB002

HIGH MOLECULAR WEIGHT HIGH DENSITY POLYETHYLENE FOR LARGE VOLUME BLOW MOLDED CONTAINER

Relene 52GB002 is a High Molecular Weight High Density Polyethylene (HMHDPE) blow molding grade with balanced stiffness and impact strength. It possesses broad molecular weight distribution, very good swell and excellent processability. 52GB002 can readily be processed on most of HDPE blow moulding machines with accumulators.

BIS Designation Code: IS 7328-3B-BB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 5 kg)	ASTM D1238	g/10 min	0.26
Melt Flow Rate (190°C/ 21.6 kg)	ASTM D1238	g/10 min	5.0
Density	ASTM D792	g/cm ³	0.952
Tensile Stress at Yield	ASTM D638	MPa	25
Elongation at Yield	ASTM D638	%	12
Flexural Modulus	ASTM D790A	MPa	1050
Notched Izod Impact Strength	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D648	°C	126

* Typical values, not to be taken as specification. All the mechanical properties are done on Compression molded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Blow moulded containers above 100L.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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52GB003

HIGH MOLECULAR WEIGHT

HIGH DENSITY POLYETHYLENE

FOR MEDIUM VOLUME BLOW MOLDED CONTAINERS

Relene 52GB003 is a High Molecular Weight High Density Polyethylene blow moulding grade with high stiffness and impact strength and good ESCR. It possesses medium molecular weight distribution, very good swell and easy processability. 52GB003 can readily be processed on most HDPE blow moulding machines.

BIS Designation Code: IS 7328-3B-BB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 5 kg)	ASTM D1238	g/10 min	0.45
Melt Flow Rate (190°C/ 21.6 kg)	ASTM D1238	g/10 min	8.5
Density	ASTM D792	g/cm ³	0.952
Tensile Stress at Yield	ASTM D638	MPa	29
Elongation at Yield	ASTM D638	%	11
Flexural Modulus	ASTM D790A	MPa	1000
Notched Izod Impact Strength	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D648	°C	125

* Typical values, not to be taken as specification. All the mechanical properties are done on Compression molded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Medium volume blow moulded containers, liners, sheets.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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54GB012

HIGH DENSITY POLYETHYLENE

FOR SMALL VOLUME BLOW MOULDED CONTAINERS

Relene 54GB012 is a High Density Polyethylene (HDPE) blow moulding grade. It possesses relatively higher density and stiffness, good flow and impact strength, good ECSR and bimodal molecular weight distribution for easy processability. 54GB012 can readily be processed on most HDPE blow moulding lines.

BIS Designation Code: IS 7328-3B-BB-FTA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 5 kg)	ASTM D1238	g/10 min	1.2
Density	ASTM D792	g/cm ³	0.954
Tensile Stress at Yield	ASTM D638	MPa	27
Elongation at Yield	ASTM D638	%	11
Flexural Modulus	ASTM D790A	MPa	900
Notched Izod Impact Strength	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D648	°C	127

*Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Edible oil, cosmetic, personal care products and general purpose blow moulding containers, Jerry cans, DWC Pipes.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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54GC016

HIGH DENSITY POLYETHYLENE

FOR CAPS AND CLOSURE

Relene 54GC016 is a bimodal High-Density Polyethylene grade suitable for injection and extrusion-compression molded Caps & Closures. This grade offers excellent process ability & organoleptic properties, very good balance of stiffness & toughness along with ESCR.

BIS Designation Code: IS 7328-3B-CB-FDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.6
Density	ASTM D792	g/cm ³	0.954
Tensile Stress at Yield	ASTM D638	MPa	27
Elongation at Yield	ASTM D638	%	9
Elongation at Break	ASTM D638	%	>900
Flexural Modulus	ASTM D790A	MPa	1200
Hardness	ASTM D2250	Shore D	62
Vicat Softening Point	ASTM D648	°C	126

*Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Caps & Closures for Still Beverages, Food & Industrial Packaging, Shoulder for Collapsible Tubes/Lamitubes.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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54GC017

HIGH DENSITY POLYETHYLENE FOR CAPS AND CLOSURE

Relene 54GC017 is a bimodal High-Density Polyethylene grade containing slip additive suitable for injection and extrusion-compression molded Caps & Closures. This grade offers excellent processability & organoleptic properties, very good balance of stiffness & toughness along with ESCR for Carbonated Soft Drink Closure Applications.

BIS Designation Code: IS 7328-3B-CB-FDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.25
Density	ASTM D792	g/cm ³	0.951
Tensile Stress at Yield	ASTM D638	MPa	27
Elongation at Yield	ASTM D638	%	9
Elongation at Break	ASTM D638	%	>900
Flexural Modulus	ASTM D790A	MPa	1050
Hardness	ASTM D2250	Shore D	62
ESCR	ASTM D5419	Weeks	>4
Vicat Softening Point	ASTM D648	°C	126

*Typical values, not to be taken as specification. All the mechanical properties are done on compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Caps & Closures for Beverage/Carbonated Soft Drinks, Food & Industrial Packaging, Shoulder for Collapsible Tubes/ Lamitubes.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

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2002F

HIGH MOLECULAR WEIGHT HIGH DENSITY POLYETHYLENE FOR SPECIALTY BLOW MOLDING

Relene 2002F is a high molecular weight, high-density polyethylene (HDPE) grade, typically designed for blow molded automotive fuel tank & gas cylinder applications. This grade can also be used for medium blow molding application. The grade demonstrates very high Environmental Stress Cracking Resistance (ESCR), good chemical resistance in combination with an excellent low temperature properties. Its high melt strength helps in production of multi-layer tanks by blow molding. This grade exhibits outstanding resistance particularly to ethanol blended gasoline.

BIS Designation Code: IS 7328-3B-BB-ETA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 21.6 kg)	ASTM D1238	g/10 min	5.5
Density	ASTM D792	g/cm ³	0.947
Tensile Strength at Yield	ASTM D638	MPa	24
Tensile Stress at Break	ASTM D638	MPa	39
Tensile Modulus	ASTM D638	MPa	1200
Elongation at Yield	ASTM D638	%	9
Elongation at Break	ASTM D638	%	800
Flexural Modulus	ASTM D790	MPa	880
ESCR @ 10% Igepal Soln (F ₅₀)	ASTM D1693B	Hrs	>1000
ESCR @ 100 % Igepal Soln (F ₅₀)	ASTM D1693B	Hrs	>1000

*Typical values, not to be taken as specification. All the mechanical properties are done on compression molded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

This grade is specially designed for blow molded Automotive fuel tanks, Gas cylinders, Medium blow molding containers etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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B56003

HIGH DENSITY POLYETHYLENE FOR GENERAL PURPOSE BLOW MOLDING

Relene B56003 is a High-Density Polyethylene (HDPE) grade suitable for general purpose extrusion blow moulding applications. Articles made from this grade exhibit good stiffness. The resin offers good melt strength, ESCR and impact resistance and typically used for general purpose containers and jerry cans for packaging of oil, vanaspati, etc.

BIS Designation Code: IS 7328-3B-BB-FTB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 5 kg)	ASTM D1238	g/10 min	1.5
Density	ASTM D792	g/cm ³	0.956
Tensile Stress at Yield	ASTM D638	MPa	26
Elongation at Break	ASTM D638	%	550
Flexural Modulus	ASTM D790A	MPa	900
Hardness	ASTM D2240	Shore D	69
Vicat Softening Point	ASTM D648	°C	128

*Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 Type IV compression moulded specimen in accordance with ASTM D4976

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

General purpose containers, jerry cans up to 20 L for packaging of oil, vanaspati, etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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E52009

HIGH DENSITY POLYETHYLENE FOR ORIENTED TAPE & MONOFILAMENT

Relene E52009 is a High-Density Polyethylene (HDPE) grade with low gel content, excellent orientation characteristic and negligible water carry-over. This grade is recommended for making raffia tapes and monofilaments (Diameter >0.20 mm) with excellent balance of high tenacity and good elongation.

BIS Designation Code: IS 7328- 3B-TB-FDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 5 kg)	ASTM D1238	g/10 min	0.90
Density	ASTM D792	g/cm ³	0.952
Tensile Stress at Yield (MD / TD)	ASTM D882	MPa	35 / 38
Ultimate Tensile Stress (MD / TD)	ASTM D882	MPa	50 / 45
Elongation at Break (MD / TD)	ASTM D882	%	900 / 950
Tear Strength (MD / TD)	ASTM D1922	g/μm	0.6 / 1.7

* Typical values, not to be taken as specification. Mechanical properties done on 40μ film with 0.75mm die gap and BUR of 2.75

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Oriented tapes, woven sacks and monofilaments.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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F46003

HIGH DENSITY POLYETHYLENE

FOR BLOW MOULDING & BLOWN FILM

Relene F46003 is a High Density Polyethylene grade suitable for general purpose blown film & extrusion blow moulding applications. Articles blown from this grade exhibit moderate to good stiffness. The resin offers good melt strength, ESCR and impact resistance.

BIS Designation Code: IS 7328-3B-FB-EDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.38
Density	ASTM D792	g/cm ³	0.946
Tensile Stress at Yield (MD / TD)	ASTM D882	MPa	29 / 26
Ultimate Tensile Stress (MD / TD)	ASTM D882	MPa	45 / 40
Elongation at Break (MD / TD)	ASTM D882	%	780 / 950
Dart Impact Strength (38 mm Dart, 66 cm height)	ASTM D1709	g/μm	1.8
Tear Strength (MD / TD)	ASTM D1922	g/μm	0.4 / 4.5

* Typical values, not to be taken as specification. Mechanical properties done on 40 μ film made with 0.8mm die gap & 2.7 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Used for producing blown film in carry bags and co-extruded film structures, general purpose blow molded containers.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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F60003

HIGH DENSITY POLYETHYLENE

FOR BLOWN & MDO FILMS

Relene F60003 is a High Density Polyethylene grade suitable for use in blends in coextruded oriented film application. This grade is designed to improve stiffness and barrier requirement in films. This resin offers good balance of mechanical and optical properties.

BIS Designation Code: IS 7328 3B-FB-GDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.7
Density	ASTM D792	g/cm ³	0.957
Tensile Stress at Yield (MD /TD)	ASTM D882	MPa	29 / 26
Ultimate Tensile Stress (MD /TD)	ASTM D882	MPa	45 / 40
Elongation at Break (MD / TD)	ASTM D882	%	780 / 950
Dart Impact Strength (38 mm Dart, 66 cm height)	ASTM D1709	g/μm	1.8
Tear Strength (MD / TD)	ASTM D1922	g/μm	0.4 / 4.5

* Typical values, not to be taken as specification. Mechanical properties done on 40 μ film made with 0.8mm die gap & 2.7 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Blown film for use in food packaging, e- commerce and other packaging applications.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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F60080

HIGH DENSITY POLYETHYLENE

FOR CAST FILM

Relene F60080 is a High Density Polyethylene (HDPE) grade suitable for film extrusion applications where higher stiffness, along with good processability is required. This grade can be used in blends with LLDPE and LDPE and can be co-extruded. Grade is designed to cater hygiene film applications where gel free films are required.

BIS Designation Code: IS 7328 3B-MB-GDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	8.0
Density	ASTM D792	g/cm ³	0.960
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	30 / 28
Tensile Modulus (MD / TD)	ASTM D882	MPa	700 / 900
Elongation at Break (MD / TD)	ASTM D882	%	850 / 800
Vicat Softening Point	ASTM D1525	°C	127
Haze	ASTM D1003	%	12

* Typical values, not to be taken as specification. Mechanical Properties are done on 25μ Cast Film.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Cast Film applications like Hygiene films

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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HD50MA180

HIGH DENSITY POLYETHYLENE

FOR INJECTION MOLDED PRODUCTS

Relene HD50MA180 is a high flow High Density Polyethylene (HDPE) grade. This grade is specially formulated to manufacture large, intricate, thin walled multi cavity injection moulded products. It has narrow molecular weight distribution, which makes it easier to process and products manufactured have good gloss and rigidity.

BIS Designation Code: IS 7328-3B-MB-EDE

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	20.0
Density	ASTM D1505	g/cm ³	0.950
Tensile Stress at Yield	ASTM D638	MPa	22
Elongation at Yield	ASTM D638	%	12
Flexural Modulus	ASTM D790	MPa	900
Notched Izod Impact Strength	ASTM D256	J/m	40
Vicat Softening Point	ASTM D1525	°C	123

* Typical values, not to be taken as specification. Mechanical Properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Houseware, storage bins, thin walled article

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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HD53EA010

HIGH DENSITY POLYETHYLENE FOR STRETCHED TAPE PRODUCTS

Relene HD53EA010 is a High Density Polyethylene (HDPE) grade produced and specially formulated to manufacture stretched tape products and other extrusion products with good combination of tenacity and elongation. It has very good processability and stretch-ability, less motor load and high output combined with good tape properties to make it a grade of choice.

BIS Designation Code: IS 7328-3B-TB-GDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.0
Density	ASTM D1505	g/cm ³	0.955
Tensile Stress at Yield (MD / TD)	ASTM D882	MPa	33 / 36
Ultimate Tensile Stress (MD / TD)	ASTM D882	MPa	48 / 45
Elongation at Break (MD / TD)	ASTM D882	%	900 / 950
Tear Strength (MD / TD)	ASTM D1922	g/μm	0.6 / 1.7

* Typical values, not to be taken as specification. Mechanical properties done on 40μ film with 0.75mm die gap and BUR of 2.75

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Oriented Tapes, Woven Sacks

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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HD55EF010

HIGH DENSITY POLYETHYLENE FOR MONOFILAMENT APPLICATION

Relene HD55EF010 is a High Density Polyethylene (HDPE) grade, specially formulated to manufacture oriented monofilament products. It has narrow molecular weight distribution and has very good processability, less motor load and high output combined with very good elongation and filament properties to make it a grade of choice.

BIS Designation Code: IS 7328-3B-LB-GDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.0
Density	ASTM D1505	g/cm ³	0.955
Tensile Stress at Yield	ASTM D638	MPa	23
Elongation at Yield	ASTM D638	%	15
Flexural Modulus	ASTM D790	MPa	820
Notched Izod Impact Strength	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D1922	°C	127

* Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Monofilaments for mosquito nets, fishing nets etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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L60075

HIGH DENSITY POLYETHYLENE FOR INJECTION MOLDING APPLICATION

Relene L60075 is a High Density Polyethylene (HDPE) grade suitable for injection moulding applications where good mechanical properties are essential with gloss, dimensional stability, good ESCR, required for technical moulding like moulded crates, luggage shells, helmets, defense applications, etc. L60075 is suitable for applications where articles are exposed to sunlight e.g. soft drink crates, defense applications, helmets, industrial pails, etc.

BIS Designation Code: IS 7328-3B-MD-FDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	8.0
Density	ASTM D1505	g/cm ³	0.960
Tensile Stress at Yield	ASTM D638	MPa	25
Elongation at Yield	ASTM D638	%	15
Flexural Modulus	ASTM D790	MPa	900
Notched Izod Impact Strength	ASTM D256	J/m	88
Hardness	ASTM D2240	Shore D	69
Vicat Softening Point	ASTM D1525	°C	128

* Typical values, not to be taken as specification. Mechanical Properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Crates, Industrial Pails and defense applications.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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M60075

HIGH DENSITY POLYETHYLENE FOR INJECTION MOLDING APPLICATION

Relene M60075 is a High - Density Polyethylene (HDPE) grade suitable for injection moulding applications where good mechanical properties are essential with gloss, dimensional stability, good ESCR, required for crates, luggage shells, helmets, defense applications, etc.

BIS Designation Code: IS 7328-3B-MB-FDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	8.0
Density	ASTM D1505	g/cm ³	0.960
Tensile Stress at Yield	ASTM D638	MPa	25
Elongation at Break	ASTM D638	%	800
Flexural Yield Strength	ASTM D790	MPa	30
Flexural Modulus	ASTM D790	MPa	900
Notched Izod Impact Strength	ASTM D256	J/m	88
Hardness	ASTM D2240	Shore D	69
Vicat Softening Point	ASTM D1525	°C	128

* Typical values, not to be taken as specification. Mechanical Properties are tested on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For Crates, Luggage and Industrial containers.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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16MA400

LOW DENSITY POLYETHYLENE

FOR INJECTION MOLDING & MASTERBATCHES

Relene 16MA400 is a Low Density Polyethylene (LDPE) with a high melt flow index for very thin, intricate, large items having adequate mechanical properties. This grade is also ideal choice for making variety of masterbatches.

BIS Designation Code: IS 7328-1-MA-BDF

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	30
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield	ASTM D638	MPa	10
Elongation at Yield	ASTM D638	%	40
Flexural Modulus	ASTM D790	MPa	140
Notched Izod Impact Strength	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D1525	°C	84

*Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Lids, Powder Coating, Injection Molding Articles, Masterbatches etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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22FA002

LOW DENSITY POLYETHYLENE FOR HEAVY DUTY FILM

Relene 22FA002 is a Low Density Polyethylene (LDPE) heavy duty film grade produced using high pressure tubular process. It has been specially designed for production of film with good Environmental Stress Crack Resistance (ESCR), mechanical and impact resistance.

BIS Designation Code: IS 7328-1-FA-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.2
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	26 / 24
Elongation at Break (MD / TD)	ASTM D882	%	400 / 600
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	5.5

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ Blown film with 0.8mm die gap and 2 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Heavy Duty Bags, Shrink Film, Canal Lining, Green House Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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24FS040

LOW DENSITY POLYETHYLENE FOR GENERAL PURPOSE FILM APPLICATION

Relene 24FS040 is a Low Density Polyethylene (LDPE) that can be used in blown film application due to excellent processability. This grade is formulated with slip additive for good surface slip and openability.

BIS Designation Code: IS 7328-1-FF-BDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.0
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	18 / 16
Elongation at Break (MD / TD)	ASTM D882	%	250 / 400
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2.0
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	>100

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Shopping bag, General Purpose Packaging etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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1005FY20

LOW DENSITY POLYETHYLENE FOR LIQUID PACKAGING

Relene 1005FY20 is a Low Density Polyethylene (LDPE) is specially designed for production of multilayered film in blend with LLDPE for packaging of liquid. This grade has been specially formulated for the production of films for liquid milk packaging on fully automatic Form Fill Seal (FFS) machines. Film produced from this grade exhibits good impact resistance and heat-sealing characteristics. The incorporation of necessary additives in the polymer ensures good surface slip, easy open ability and smooth flow of film on packaging machines.

BIS Designation Code: IS 7328-1-FF-BXDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.5
Density	ASTM D1505	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	21 / 20
Elongation at Break (MD / TD)	ASTM D882	%	300 / 525
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.7
Haze	ASTM D1003	%	8
Gloss @ 60°	ASTM D2457	-	>80

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Slip grade for Mono, Co-extruded Films for Liquid Packaging, General Purpose Consumer Product Packaging.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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1020FA20

LOW DENSITY POLYETHYLENE

FOR GENERAL PURPOSE & ADHESIVE-LAMINATION

Relene 1020FA20 is a Low Density Polyethylene (LDPE) film grade and can be processed easily on conventional blown film and cast film lines. This grade is designed to facilitate adhesion during lamination.

BIS Designation Code: IS 7328-1-VA-BXDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	2.0
Density	ASTM D1505	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	21 / 20
Elongation at Break (MD / TD)	ASTM D882	%	275 / 425
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.0
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	95

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Hygiene Film, Bubble Wrap Film, Adhesive Lamination Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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1070LA17

LOW DENSITY POLYETHYLENE FOR EXTRUSION COATING

Relene 1070LA17 is a Low Density Polyethylene (LDPE) used in extrusion coating produced using high pressure autoclave process. This grade is specially developed for extrusion coating on different substrates and contains no additives. It exhibits excellent drawdown characteristics maintaining good adhesion, with very low neck-in even at higher line speeds. It can also be easily processed into thin gauge films for adhesive lamination, heat & pressure lamination.

BIS Designation Code: IS 7328-1-HA-BXDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	8.5
Density	ASTM D1505	g/cm ³	0.917
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	18 / 15
Elongation at Break (MD / TD)	ASTM D882	%	250 / 400

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ cast film.

Typical Processing Conditions

Processing temperatures: 200 - 320 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Extrusion Coating, Heat & Pressure Lamination Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J22FA002E

LOW DENSITY POLYETHYLENE FOR HEAVY DUTY FILM

Relene J22FA002E is a Low Density Polyethylene (LDPE) heavy duty film grade produced using high pressure tubular process. It has been specially designed for production of film with good Environmental Stress Crack Resistance (ESCR), mechanical and impact resistance.

BIS Designation Code: IS 7328-1-FA-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.5
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	26 / 24
Elongation at Break (MD / TD)	ASTM D882	%	400 / 600
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	5.5

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ Blown film with 0.8mm die gap and 2 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Heavy Duty Bags, Shrink Film, Canal Lining, Green House Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J24FA040

LOW DENSITY POLYETHYLENE FOR GENERAL PURPOSE FILM APPLICATION

Relene J24FA040 is a Low Density Polyethylene (LDPE) that can be used in blown film application due to excellent processability. This grade can be used for lamination and coating application with excellent melt drawing.

BIS Designation Code: IS 7328-1-FA-BDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.0
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	18 / 16
Elongation at Break (MD / TD)	ASTM D882	%	250 / 400
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2.0
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	>100

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Shopping Bag, General Purpose Packaging, Lamination Film, in blending in Extrusion Coating etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J24FS040

LOW DENSITY POLYETHYLENE FOR GENERAL PURPOSE FILM APPLICATION

Relene J24FS040 is a Low Density Polyethylene (LDPE) that can be used in blown film application due to excellent processability. This grade is formulated with suitable additive for good surface slip and easy openability.

BIS Designation Code: IS 7328-1-FH-BDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.0
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	18 / 16
Elongation at Break (MD / TD)	ASTM D882	%	250 / 400
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2.0
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	>100

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Slip Grade for Manufacturing Film used in Shopping Bag, General Purpose Packaging, Lamination Film, Zip Lock Bags etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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J1005FS20

LOW DENSITY POLYETHYLENE FOR LIQUID PACKAGING

Relene J1005FS20 is a Low Density Polyethylene (LDPE) film grade basically used in liquid packaging application. This grade has been specially formulated for the production of films for liquid milk packaging on fully automatic Form Fill Seal (FFS) machines. Film produced from this grade exhibits good impact resistance and heat-sealing characteristics. The incorporation of necessary additives in the polymer ensures good surface slip, easy open ability and smooth flow of film on packaging machines.

BIS Designation Code: IS 7328-1-FH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.5
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	23 / 22
Elongation at Break (MD / TD)	ASTM D882	%	300 / 575
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.2
Haze	ASTM D1003	%	8
Gloss @ 60°	ASTM D2457	-	>90

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Slip Grade for Mono, Co-Extruded Films for Liquid Packaging, General Purpose Consumer Product Packaging.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J1005FY20

LOW DENSITY POLYETHYLENE FOR LIQUID PACKAGING

Relene J1005FY20 is a Low Density Polyethylene (LDPE) film grade basically used in liquid packaging application. This grade has been specially formulated for the production of films for liquid milk packaging on fully automatic Form Fill Seal (FFS) machines. Film produced from this grade exhibits good impact resistance and heat-sealing characteristics. The incorporation of necessary additives in the polymer ensures good surface slip, easy open ability and smooth flow of film on packaging machines.

BIS Designation Code: IS 7328-1-FH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.65
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	23 / 22
Elongation at Break (MD / TD)	ASTM D882	%	300 / 575
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.2
Haze	ASTM D1003	%	8
Gloss @ 60°	ASTM D2457	-	>90

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Slip Grade for Mono, Co-Extruded Films for Liquid Packaging, General Purpose Consumer Product Packaging.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J1007FA20

LOW DENSITY POLYETHYLENE

FOR FILM & FOAM APPLICATION

Relene J1007FA20 is a Low Density Polyethylene (LDPE) film grade basically used in film application. This grade is specially designed for easy processing on blown film plants. This grade does not contain any slip additive. This grade can be used for blending in foam application.

BIS Designation Code: IS 7328-1-VA-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.7
Density	ASTM D792	g/cm ³	0.923
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	21 / 20
Elongation at Break (MD / TD)	ASTM D882	%	250 / 500
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.1
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	70

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Co-Extruded Film for Lamination, blending in Foam Application.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J1020FA20

LOW DENSITY POLYETHYLENE

FOR GENERAL PURPOSE & ADHESIVE-LAMINATION

Relene J1020FA20 is a Low Density Polyethylene (LDPE) film grade and can be processed easily on conventional blown film and cast film lines. This grade is designed to facilitate adhesion during lamination.

BIS Designation Code: IS 7328-1-KA-BDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	2.0
Density	ASTM D792	g/cm ³	0.921
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	21 / 20
Elongation at Break (MD / TD)	ASTM D882	%	275 / 425
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2.0
Haze	ASTM D1003	%	7
Gloss @ 60°	ASTM D2457	-	95

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 0.8mm die gap and 2 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Adhesive Lamination, Hygiene Film, Bubble Wrap Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J1020FF20

LOW DENSITY POLYETHYLENE FOR FOAM APPLICATION

Relene J1020FF20 is a Low Density Polyethylene (LDPE) used in foam as well as film applications. This grade has very good melt strength which provide uniform cell structure, cell growth and stiffness in the foam. This grade can be used in making foam using both physical and chemical blowing agent.

BIS Designation Code: IS 7328-1-VA-BDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.8
Density	ASTM D792	g/cm ³	0.924
Tensile Strength at Yield	ASTM D638	MPa	10
Tensile Modulus	ASTM D638	MPa	350
Elongation at Break	ASTM D638	%	400
Hardness	ASTM D2240	Shore D	45
Vicat Softening Point	ASTM D1525	°C	96

*Typical values, not to be taken as specification. All mechanical properties are done on Compression Molded specimen in accordance to ASTM D4703

Typical Processing Conditions

Processing temperatures: 150 - 200 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Foam for Mattress, Automotive, Footwear and Industrial Application, Adhesive Lamination Film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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J1020XA20

LOW DENSITY POLYETHYLENE

FOR W&C APPLICATION

Relene J1020XA20 is a Low Density Polyethylene (LDPE) used as base resin for manufacturing of medium voltage cable insulation with suitable doses of cross linking agents. This grade is designed for easy processing on conventional cable extrusion line. It does not contain any additives. Required amount of antioxidant must be added to achieve desired ageing performance.

BIS Designation Code: IS 7328-1-JA-BDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	2.0
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield	ASTM D638	MPa	8.5
Ultimate Tensile Strength	ASTM D638	MPa	12
Tensile Modulus	ASTM D638	MPa	170
Elongation at Break	ASTM D638	%	>500
Hardness	ASTM D2240	Shore D	45
Vicat Softening Point	ASTM D1525	°C	89

*Typical values, not to be taken as specification. All mechanical properties are done on Compression Molded specimen in accordance to ASTM D4703

Typical Processing Conditions

Processing temperatures: 150 - 200 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Medium Voltage Cable Insulation application.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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E24065

LINEAR LOW DENSITY POLYETHYLENE FOR EXTRUSION COATING

Relene E24065 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), and designed for extrusion coating applications in blends with LDPE.

BIS Designation Code: IS 7328-2B-HB-BDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	6.5
Density	ASTM D792	g/cm ³	0.924
Tensile Strength at Yield	ASTM D638	MPa	11
Ultimate Tensile Strength	ASTM D638	MPa	15
Elongation at Break	ASTM D638	%	740
Flexural Modulus	ASTM D790	MPa	250
Notch Impact Strength (23°C)	ASTM D256	J/m	No Break
Vicat Softening Point	ASTM D1532	°C	102

* Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Extrusion coating on Woven Fabric, Jute Fabric, Al Foil etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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F17030

LINEAR LOW DENSITY POLYETHYLENE FOR CAST FILM EXTRUSION

Relene F17030 is a butene co-monomer based Linear Low Density Polyethylene grade with optimum level of Antioxidants. This grade is designed for good processing on Cast Film lines with balanced mechanical & optical properties.

BIS Designation Code: IS 7328-2B-KBE-BDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	3
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	7 / 8
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	22 / 20
Elongation at Break (MD / TD)	ASTM D882	%	500 / 600
Tear Strength (MD / TD)	ASTM D1922	g/μm	2 / 5
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Haze	ASTM D1003	%	2
Gloss @ 60°	ASTM D2457	-	95

* Typical values, not to be taken as specification. Mechanical properties are done on 40μ cast film made with 0.8mm die gap

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Co-Ex films, Stretch & Cling films, Hygiene films, Personal Care & General Purpose packaging films.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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F18010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene F18010 is a butene co-monomer based Linear Low Density Polyethylene grade with optimum level of anti-oxidants designed for blown film applications. They have excellent draw down characteristics with very good impact strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBE-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	12.5 / 12
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	38 / 30
Elongation at Break (MD / TD)	ASTM D882	%	700 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	3.3 / 10.3
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.5
Haze	ASTM D1003	%	14

* Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Blown film for lamination film, liquid packaging, surface protection film, stretch cling film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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F19010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene F19010 is a butene co-monomer based Linear Low Density Polyethylene grade with optimum levels of slip, anti-block and antioxidants. The grades are designed for blown film applications. They have excellent draw down characteristics with very good impact strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.0
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	12.5 / 12
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	38 / 30
Elongation at Break (MD / TD)	ASTM D882	%	700 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	3.3 / 10.3
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.5
Haze	ASTM D1003	%	14

* Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Blown film for liner bags, lamination film, liquid packaging, consumer product packaging etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

JD22010

LINEAR LOW DENSITY POLYETHYLENE FOR DRIP LATERALS

Relene JD22010 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), for Drip lateral applications. Drip laterals produced using this grade have outstanding processability, tensile and elongation properties. This grade meets the requirements of material as per IS 12786: 2024 on 'Irrigation equipment-Polyethylene pipes for irrigation laterals specification'. When incorporated with adequate quantity of carbon black as specified in clause 4.1 of IS 12786: 2024, the pipe made from this grade will meet the creep rupture stress requirement of PE-25 class.

BIS Designation Code: IS 7328-2B-NB-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.1
Density	ASTM D792	g/cm ³	0.922
Tensile Strength at Break	ASTM D638	MPa	20
Elongation at Break	ASTM D638	%	1000
Environmental Stress Crack (F ₅₀ , 10% Igepal, 50°C)	ASTM D1693	Hr	>1000
Haze	ASTM D1003	%	14

* Typical values, not to be taken as specification. Mechanical properties are done on Compression Molded Specimen in accordance with ASTM D4976.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For flat and round drip laterals

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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JF18010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene JF18010 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants. It does not contain slip and anti-block. This grade is designed for flexible packaging applications. It has excellent draw down characteristics with very good toughness, tensile strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBE-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	13 / 14
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	50 / 36
Elongation at Break (MD / TD)	ASTM D882	%	800 / 1000
Tear Strength (MD / TD)	ASTM D1922	g/μm	5 / 15
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.5
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Haze	ASTM D1003	%	13
Gloss @ 60°	ASTM D2457	-	75

*Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For lamination film, liquid packaging, surface protection films, stretch cling films, etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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JF18020

LINEAR LOW DENSITY POLYETHYLENE FOR FILM EXTRUSION, WIRE & CABLE APPLICATION

Relene JF18020 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants. It does not contain slip and anti-block and is designed for good processing on blown and cast film lines with excellent balance in mechanical and optical properties. It is also used as base resin for wire and cable compounds using silane crosslinking process.

BIS Designation Code: IS 7328-2B-KBE-BDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	2.3
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at break (MD / TD)	ASTM D882	MPa	38 / 20
Elongation at Break (MD / TD)	ASTM D882	%	500 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	1.8 / 16
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	1.9
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Haze	ASTM D1003	%	0.8
Gloss @ 20°	ASTM D 2457	-	105

*Typical values, not to be taken as specification. Mechanical properties are done on 25 μ cast film made with 0.8mm die gap.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Films for industrial packaging, stretch wrap, cling film for food & hygiene packaging, protective films, bubble wrap, Wire & Cable application etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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JF18030

LINEAR LOW DENSITY POLYETHYLENE FOR W&C COMPOUND AND CAST FILM

Relene JF18030 is a butene comonomer-based Linear Low Density Polyethylene (LLDPE) for Cast Film & base resin for wire and cable compounds using silane crosslinking process. This grade can be used to produce high strength film. The grade does not contain slip and anti-block.

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	3.2
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at break (MD / TD)	ASTM D882	MPa	30 / 25
Elongation at Break (MD / TD)	ASTM D882	%	650 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	2.5 / 11
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2.5
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Haze	ASTM D1003	%	2.0
Gloss @ 60°	ASTM D 2457	-	150

*Typical values, not to be taken as specification. Mechanical properties are done on 25 μ cast film made with 0.8 mm die gap.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Wire and Cable Compounds & Cast Film Applications

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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JF19010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene JF19010 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants, slip and anti-block. This grade is designed for blown film applications. It has excellent draw down characteristics with very good toughness, tensile strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1.1
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	13 / 14
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	50 / 36
Elongation at Break (MD / TD)	ASTM D882	%	800 / 1000
Tear Strength (MD / TD)	ASTM D1922	g/μm	5 / 15
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3.5
Haze	ASTM D1003	%	13
Gloss @ 60°	ASTM D 2457	-	75

*Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For liners, lamination film, liquid packaging, consumer packaging films, agricultural films etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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JF19020

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene JF19020 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants, slip and anti-block designed for good processing on blown film and cast film with balanced mechanical and optical properties.

BIS Designation Code: IS 7328-2B-FBH-BDB

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	2
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	12 / 10
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	36 / 28
Elongation at Break (MD / TD)	ASTM D882	%	800 / 950
Tear Strength (MD / TD)	ASTM D1922	g/μm	2.5 / 7.5
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	2
Haze	ASTM D1003	%	15
Gloss @ 60°	ASTM D 2457	-	65

*Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For co-extrusion, agricultural films, shopping bags, general purpose packaging films etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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JLL36RA045

LINEAR LOW DENSITY POLYETHYLENE FOR ROTATIONAL MOULDING APPLICATION

Relene JLL36RA045 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE) with optimum level of antioxidants. This grade when blended with appropriate amount of carbon black can withstand continuous outdoor exposure. JLL36RA045 when pulverized into 20 – 30 mesh powder offers good processability. The grade has very good Environmental Stress Cracking Resistance (ESCR) and balance of stiffness and impact strength.

BIS Designation Code: IS 7328-2B-RB-CDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/cm ³	0.936
Tensile Strength at Yield	ASTM D638	MPa	18
Elongation at Break	ASTM D638	%	>750
Flexural Modulus	ASTM D790	MPa	700
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR (F ₅₀ , 10% Igepal)	ASTM D1693	Hr	>350
Hardness	ASTM D2240	Shore D	50
Vicat Softening Point	ASTM D1525	°C	114

*Typical values, not to be taken as specification. Mechanical Test are done on Compression Molded Specimen with accordance to ASTM D4976.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Over-head water tanks, loft tanks, litter bins, toys etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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JLL36RA045UV

LINEAR LOW DENSITY POLYETHYLENE FOR ROTATIONAL MOULDING APPLICATION

Relene JLL36RA045UV is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE) with optimum level of antioxidant and UV stabilizer. This grade is stabilized with appropriate quantity of HALS to give a long service with continuous outdoor exposure with or without blending of any carbon black. JLL36RA045UV when pulverized into 20-30 mesh powder offers good processability. This grade has very good Environmental Stress Crack Resistance (ESCR) with a balance of impact and stiffness.

BIS Designation Code: IS 7328-2B-RD-CDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/cm ³	0.936
Tensile Strength at Yield	ASTM D638	MPa	18
Elongation at Break	ASTM D638	%	>750
Flexural Modulus	ASTM D790	MPa	700
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR (F ₅₀ , 10% Igepal)	ASTM D1693	Hr	>350
Hardness	ASTM D2240	Shore D	50
Vicat Softening Point	ASTM D1525	°C	114

*Typical values, not to be taken as specification. Mechanical Test are done on Compression Molded Specimen with accordance to ASTM D4976.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Rotational molded products used in outdoor application like water tanks, road blocks etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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LL20FA010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene LL20FA010 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants. It does not contain slip and anti-block. This grade is designed for flexible packaging applications. It has excellent draw down characteristics with very good toughness, tensile strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBE-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	40 / 30
Elongation at Break (MD / TD)	ASTM D882	%	700 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	2.5 / 7.4
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5

*Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Lamination film, Surface protection film, Stretch Cling film etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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LL20FS010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM EXTRUSION

Relene LL20FS010 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum levels of antioxidants, slip and anti-block. This grade is designed for blown film applications. It has excellent draw down characteristics with very good toughness, tensile strength and tear resistance.

BIS Designation Code: IS 7328-2B-FBH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	1
Density	ASTM D792	g/cm ³	0.920
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	40 / 30
Elongation at Break (MD / TD)	ASTM D882	%	700 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	3.3 / 10.3
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709A	g/μm	3

*Typical values, not to be taken as specification. Mechanical properties are done on 40 μ blown film made with 1.8mm die gap & 2.5 BUR

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

For liners, lamination film, liquid packaging, consumer packaging films, agricultural films etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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LL36RA045

LINEAR LOW DENSITY POLYETHYLENE FOR ROTATIONAL MOULDING APPLICATION

Relene LL36RA045 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE). This grade when blended with appropriate amount of carbon black can withstand continuous outdoor exposure. LL36RA045 when pulverized into 20 – 30 mesh powder offers good processability. The grade has very good Environmental Stress Cracking Resistance (ESCR) and a combination of stiffness and impact strength.

BIS Designation Code: IS 7328-2B-RB-CDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/cm ³	0.936
Tensile Strength at Yield	ASTM D638	MPa	16
Elongation at Yield	ASTM D638	%	14
Flexural Modulus	ASTM D790	MPa	700
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR (F ₅₀ , 10% Igepal)	ASTM D1693	Hr	>350
Hardness	ASTM D2240	Shore D	55
Vicat Softening Point	ASTM D1525	°C	117

*Typical values, not to be taken as specification. Mechanical Tests are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Over-head water tanks, loft tanks, litter bins etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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LL36RA045UV

LINEAR LOW DENSITY POLYETHYLENE FOR ROTATIONAL MOULDING APPLICATION

Relene LL36RA045UV is a Butene co-monomer based Linear Medium Density Polyethylene (LLDPE). This grade is stabilized with appropriate quantity of HALS to give a long service with continuous outdoor exposure with or without blending of any carbon black. LL36RA045UV when pulverized into 20-30 mesh powder offers good processability. This grade has very good Environmental Stress Crack Resistance (ESCR) and a combination of stiffness and impact strength.

BIS Designation Code: IS 7328-2B-RD-CDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/cm ³	0.936
Tensile Strength at Yield	ASTM D638	MPa	18
Elongation at Yield	ASTM D638	%	20
Flexural Modulus	ASTM D790	MPa	700
Notched Izod Impact Strength	ASTM D256	J/m	No Break
ESCR (F ₅₀ , 10% Igepal)	ASTM D1693	Hr	>350
Hardness	ASTM D2240	Shore D	55
Vicat Softening Point	ASTM D1525	°C	117

*Typical values, not to be taken as specification. Mechanical Test are done on Compression Molded Specimen with accordance to ASTM D4976.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Rotational molded products used in outdoor application like water tanks, road blocks etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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M24200

LINEAR LOW DENSITY POLYETHYLENE

FOR INJECTION MOLDING, MASTERBATCHES & COMPOUNDS

Relene M24200 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with narrow molecular weight distribution and optimum level of antioxidants. It offers excellent processability and is recommended as base resin for preparing masterbatch. It can also be used for producing injection moulded lids, containers, houseware & general purpose articles as this grade exhibits good flexibility and low warpage.

BIS Designation Code: IS 7328-2B-MB-BDE

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	20
Density	ASTM D792	g/cm ³	0.924
Tensile Strength at Yield	ASTM D638	MPa	10
Elongation at Break	ASTM D638	%	>600
Flexural Modulus	ASTM D790	MPa	255
Vicat Softening Point	ASTM D 1525	°C	96

*Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Masterbatches, Housewares, Lids, Shopping Baskets, etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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M26500

LINEAR LOW DENSITY POLYETHYLENE

FOR INJECTION MOLDING, MASTERBATCHES & COMPOUNDS

Relene M26500 is a Butene co-monomer based Linear Low Density Polyethylene (LLDPE), with narrow molecular weight distribution and optimum level of antioxidants. It offers excellent processability and is recommended as base resin for preparing masterbatch. It can also be used for producing injection moulded lids, containers, houseware & general purpose articles as this grade exhibits good flexibility and low warpage.

BIS Designation Code: IS 7328-2B-MB-BDG

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	50
Density	ASTM D792	g/cm ³	0.926
Tensile Strength at Yield	ASTM D638	MPa	11.5
Elongation at Break	ASTM D638	%	600
Flexural Modulus	ASTM D790	MPa	250
Vicat Softening Point	ASTM D 1525	°C	97

*Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Masterbatches, Housewares, Lids, Shopping Baskets, etc.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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O19010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM APPLICATION

Relene O19010 is an Octene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum level of antioxidants, slip and anti-block. This grade is designed to make blown films for heavy duty applications and liquid packaging where excellent sealing is required. This grade exhibits excellent draw down characteristics, impact strength, toughness and tear resistance.

BIS Designation Code: IS 7328-20-FBH-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.9
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	12.5 / 13
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	40 / 35
Elongation at Break (MD / TD)	ASTM D882	%	650 / 800
Tear Strength (MD / TD)	ASTM D1922	g/μm	13.8 / 27.6
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	7
Gloss @ 60°	ASTM D 523	%	80

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 1.8mm die gap and 2.5 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Specialty film for liquid packaging

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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O21010

LINEAR LOW DENSITY POLYETHYLENE FOR BLOWN FILM APPLICATION

Relene O21010 is an Octene co-monomer based Linear Low Density Polyethylene (LLDPE), with optimum level of antioxidants and polymer processing aids. It does not contain slip and anti-block. This grade is designed for lamination film applications where good optical properties and adhesion to the substrate are required.

BIS Designation Code: IS 7328-20-FBE-BDA

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	0.9
Density	ASTM D792	g/cm ³	0.918
Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	12.5 / 13
Ultimate Tensile Strength (MD / TD)	ASTM D882	MPa	45 / 40
Elongation at Break (MD / TD)	ASTM D882	%	750 / 950
Tear Strength (MD / TD)	ASTM D1922	g/μm	13.8 / 27.6
Dart Impact Strength, F ₅₀ (38mm dart, 66cm height)	ASTM D1709	g/μm	7
Co-efficient of Friction (COF)	ASTM D1894	-	>0.5
Gloss @ 60°	ASTM D 523	%	80

*Typical values, not to be taken as specification. Mechanical properties are done on 40μ blown film made with 1.8mm die gap and 2.5 BUR.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Specialty Lamination Film

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

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O35042

LINEAR LOW DENSITY POLYETHYLENE FOR ROTATIONAL MOULDING APPLICATION

Relene O35042 is an Octene co-monomer based Linear Low Density Polyethylene (LLDPE), designed for rotational molding application. This grade can be used for producing special products like automotive component, manholes, diesel fuel tanks, insulated fish crates, and pallets. Due to presence of octene co-monomer and medium density of the polymer, it exhibits excellent toughness, impact resistance and Environmental Stress Crack Resistance (ESCR).

BIS Designation Code: IS 7328-20-RB-CDC

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/cm ³	0.933
Tensile Strength at Yield	ASTM D638	MPa	15
Ultimate Tensile Strength	ASTM D638	MPa	27
Elongation at Break	ASTM D638	%	950
Flexural Modulus	ASTM D790	MPa	500
Izod Impact Strength	ASTM D256	J/m	No Break
Hardness	ASTM D2240	Shore D	57
Vicat Softening Point	ASTM D256	°C	115

*Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Diesel fuel tanks, chemical storage tanks, septic tanks, general purpose tanks.

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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X24065

LINEAR LOW DENSITY POLYETHYLENE FOR MEDIUM VOLTAGE XLPE CABLE

Relene X24065, is a Butene co-monomer based Linear Low Density Polyethylene. This grade is designed for XLPE (silane grafted) cable compounds. It offers superior Processability, ESCR and Abrasion resistance.

BIS Designation Code: IS 7328-2B-JB-BDD

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (190°C/ 2.16 kg)	ASTM D1238	g/10 min	6.4
Density	ASTM D792	g/cm ³	0.924
Tensile Strength at Yield	ASTM D638	MPa	11
Tensile Strength at Break	ASTM D638	MPa	15
Elongation at Yield	ASTM D638	%	17
Elongation at Break	ASTM D638	%	>600
Flexural Modulus	ASTM D790	MPa	260
Hardness	ASTM D2240	Shore D	53

* Typical values, not to be taken as specification. Mechanical properties are done on Injection Molded Specimen.

Typical Processing Conditions

Processing temperatures: 180 - 220 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Medium voltage XLPE cable compounding application

Regulatory Information

The product complies with Indian Standard IS 10146 on “Specification for Polyethylene for safe use in contact with foodstuffs, pharmaceuticals, and drinking water.” It also conforms to IS 16738:2018 on “Positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals.”

The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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5401P

RANDOM COPOLYMER

FOR EBM, 1-stage ISBM, BFS-IV bottles

Relpure 5401P is a random copolymer polypropylene grade with low melt flow, offering excellent processability, good clarity and impact resistance. It is specifically designed for applications in blow molding and extrusion. **Relpure 5401P** is well-suited for pharmaceutical, cosmetic, and food-contact products and is also suitable for medical application.

BIS Designation Code: IS 10951-2-BB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	1.9
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	26
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	800
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	65
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	90
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 180 – 240 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

BFS IV Bottles, ISBM IV Bottles, Cosmetic containers and Food Containers.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. This grade comply with FDA : CFR Title 21,177.1520, US Pharmacopoeia test(USP class VI), USP 661.1, IP 6.2.1.1.1 as well as the European Pharmacopoeia test(EP 3.1.6).

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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The product(s) may not be used in:

- (i) U.S. FDA Class III Medical Devices; Health Canada Class IV Medical Devices; European Class III Medical Devices.*
- (ii) applications involving permanent implantation into the body;*
- (iii) life-sustaining medical applications.*

All references to U.S. FDA, Health Canada, and European Union regulations include another country's equivalent regulatory classification.



POLYPROPYLENE (PP) - Grades & Applications

Reliance Polymers offers a wide range of Homopolymer, Random Copolymer and Impact Copolymer grades in Polypropylene.

Homopolymer grades are available in a wide range of melt flows. They are carefully designed to suit the requirements of various applications like blow and injection moulding, BOPP films, woven sacks, FIBC and fibres.

Random Copolymer grades are designed keeping in mind the critical requirements of pipes and fittings and clarified applications in blow and injection moulding and ISBM.

Impact Copolymers are tailored to meet the requirements of various applications like furniture, paint pails, caps and closures, luggage, crates, appliances and compounding.

HOMOPOLYMER		
Grade	MFI g/10 min	Typical Applications
FILMS		
H029SG	2.9	General purpose BOPP film
H034SG	3.4	Plain & Metallisable BOPP film
H080EY	8.0	Textile over wraps, Garment bags, Snack food packaging
H100EY	10.0	
WOVEN SACKS & MONOFILAMENTS		
H026SG	2.6	High tenacity tapes FIBC, Ropes & twines
H030SG	3.4	Woven fabrics, Sacks & Monofilaments
AS030N	3.0	
SS35N	3.0	
EXTRUSION COATING		
H350EG	38.0	Extrusion Coating, Lamination
FIBRE & FILAMENT		
AS160N	16.0	Staple fibre & Multifilaments
H200FG	20.0	Multifilament yarn, Staple fibre
H230FG	25.0	BCF applications
H350FG	38.0	Fine fibre yarn, Non-woven fabric
EXTRUSION AND BLOW MOULDING		
H020EG	2.0	Sheets, Strappings, Tubings, Blow moulded containers
INJECTION MOULDING		
H050MN	5.0	Furniture & Rigid packaging
H110MA	10.0	Rigid packaging, Furniture & Housewares
AM120N	12.0	Industrial mouldings & Furniture
H200MA	25.0	Thin wall containers, Multicavity moulding
H200MK	20.0	Clear thin wall containers, Multicavity moulding, Syringes
AM650N	65.0	Thin walled injection moulded articles, Compounding

RANDOM COPOLYMER		
Grade	MFI g/10 min	Typical Applications
EXTRUSION AND BLOW MOULDING		
AER003N	0.3	Pipes, Fittings, Sheets & Thermoforming
SR20NC	1.9	High clarity bottles, Containers, Sheets
SR20NS	1.9	Blow moulding, Extrusion
INJECTION MOULDING		
R120MK	12.0	Clarified Random Copolymer - Rigid containers, Housewares, ISBM bottles, Syringes
SRM100NC	12.0	
T200MK	20.0	
SRM250NC	25.0	
IMPACT COPOLYMER		
EXTRUSION AND BLOW MOULDING		
C015EG	1.5	Sheets, Blow moulded containers
EXTRUSION COATING		
B250EG	25.0	Extrusion coating
INJECTION MOULDING		
B030MG	3.0	Furniture, Industrial products
AMI035N	3.5	
MI3530	3.5	
MI3535	3.5	Caps, Closures
C080MA	8.0	Furniture, Industrial products
C080MT	8.0	Batteries, Automotive
B120MA	12.0	Housewares, General purpose moulding
C120MN	12.0	Paint pails, Luggage, Industrial components
D120MA	12.0	Automotive compounds
C200MN	20.0	Collapsible crates, Automotive compounds
B220MN	22.0	Appliances, Automotive
C320MN	32.0	Appliances, Automotive, Compounding
B400MN	35.0	Appliances, Automotive, Compounding
AMI400N	40.0	Thin wall moulding, Compounding
B550MN	55.0	Appliances, Compounding
B650MN	65.0	Appliances, Automotive, Compounding

Note: MFI (230°C/2.16 kg) g/10 min ASTM D1238



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AM120N

HOMOPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol AM120N is PP homopolymer grade recommended for Injection Molding. It is an ideal choice for general purpose injection molding and suitable for making furniture compounds.

BIS Designation Code: IS 10951-1-MB-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	37
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1450
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 260 °C

Mold temperature: 30 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Industrial moldings & Furniture Compounds.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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AM650N

HOMOPOLYMER

FOR INJECTION MOULDED PRODUCTS

Repol AM650N is PP homopolymer grade recommended for use in Injection Molding. It is a material of choice for use in Compounding / Masterbatch.

BIS Designation Code: IS 10951-1-MB-F

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	65
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1350
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	25
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 260 °C

Mold temperature: 20 – 40 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

TWIM containers, Articles with intricate long flow paths and Multicavity molding, Compounding & Masterbatch Manufacturing.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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AS030N

HOMOPOLYMER

FOR WOVEN SACKS AND MONOFILAMENTS

Repol AS030N is a PP homopolymer grade recommended for raffia tapes used in plastics woven sack. It exhibits good processability and excellent mechanical properties. The grade is designed for low water carry over.

BIS Designation Code: IS 10951-1-TB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1400
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	44
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 220 – 260 °C

Water bath temperatures: 30 – 50 °C

Stretching temperatures: ~ 155 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Woven fabrics, Sacks & Monofilaments, Carpet backing, Industrial packaging and Geotextiles.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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AS160N

HOMOPOLYMER

FOR MULTIFILAMENTS & STAPLE FIBER

Repol AS160N is PP homopolymer grade recommended for multifilament yarn and staple fiber. It has excellent processability for production of high tenacity multifilament yarn.

BIS Designation Code: IS 10951-1-LB-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	16
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	37
Elongation at Yield (50 mm/min.)	ASTM D638	%	11
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1500
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	25
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 210 – 260 °C
Quenching Temperature: 10 – 20 °C
Godet rolls Temperature: 80 – 120 °C
Draw Ratio: ~ 3.2
Texturizing Temperature: 125 – 165°C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Multifilament yarn, staple fiber etc.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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AS12000N

HOMOPOLYMER

FOR MELT BLOWN NON-WOVEN

Repol AS12000N is an ultra-high melt flow, narrow molecular weight distribution polypropylene homopolymer grade used for Melt Blown Non-Woven. High Melt Flow and Narrow MWD helps in optimum processing / line speeds and can be melt blown in to ultra-fine denier fibers.

BIS Designation Code: IS 10951-1-CB-G

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230 °C/2.16 kg) ^{\$}	ASTM D1238	g/10 min	1200
Density	ASTM D792	g/cc	0.90
Melting Temperature	ASTM D3418	°C	161 - 165

* Typical values, not to be taken as specification.

Typical Processing Conditions

Extrusion Temperature: 260 – 290 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Melt blown nonwoven fabrics for Absorption & Filtration, Hygiene & Medical Disposables, Masks, Wipes and others.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with the FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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H019TG

HOMOPOLYMER

FOR EXTRUSION AND BLOW MOLDED PRODUCTS

Repol H019TG is Polypropylene Homopolymer grade recommended for use in Extrusion & Blow Molding applications. It is an ideal material to be used for strappings, stationery items etc.

BIS Designation Code: IS 10951-1-DB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	1.9
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1700
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	40
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 170 – 250 °C

Applications

Sheets, strappings, tubing, blow molded containers, straws, ball pen refills, sheets and thermoformed containers.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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H026SG

HOMOPOLYMER

FOR FIBC & GEOTEXTILES

Repol H026SG is a PP homopolymer grade recommended for making raffia tapes used in making plastic woven sack. It exhibits good processability and excellent mechanical properties. The grade is formulated with additive package for low water carry over.

BIS Designation Code: IS 10951-1-TB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	2.6
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1700
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	40
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 220 – 260 °C

Water bath temperatures: 30 – 50 °C

Stretching temperatures: ~ 155 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Woven fabrics for FIBC, Geotextiles, Ropes, Monofilaments & carpet backing.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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H030SG

HOMOPOLYMER

FOR WOVEN SACKS & MONOFILAMENTS

Repol H030SG is a PP homopolymer grade recommended for making raffia tapes used in plastics woven sack. It exhibits good processability and excellent mechanical properties. The grade is designed for low water carry over.

BIS Designation Code: IS 10951-1-TB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3.4
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1650
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	40
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 220 – 260 °C

Water bath temperatures: 30 – 50 °C

Stretching temperatures: ~ 155 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Woven fabrics, Sacks & Monofilaments, Carpet backing, Industrial packaging and Geotextiles.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

H034SG

HOMOPOLYMER FOR BOPP FILMS

Repol H034SG is PP homopolymer grade recommended for use in Oriented Film process. It is suitable for low thickness films and processing on high line speeds. It is the material of choice for BOPP films used in adhesive tapes, food packaging, cigarette & textile overwraps, synthetic paper, etc.

BIS Designation Code: IS 10951-1-FB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	gm/10 min	3.4
Density	ASTM D792	g/ cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	36
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1350
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	40
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	69

* Typical values. Not to be taken as specifications. All the mechanical properties as per ASTM D638 injection molded Type I specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 220 – 260 °C

Chill roll temperatures: 20 – 50 °C

Stretching temperatures: MDO: 120 – 160 °C; TDO: 150 – 170 °C

Stretch ratio: MDO: 4-8; TDO: 7-10

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Plain & metallisable BOPP films, Adhesive tapes, Food packaging, Cigarette & Textile Overwraps, Synthetic Paper

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA: CFR Title 21,177.1520, Olefin polymers

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H045SG

HOMOPOLYMER

FOR WOVEN SACKS & MONOFILAMENTS

Repol H045SG is PP homopolymer grade recommended for making raffia tapes used in making plastic woven sack. The grade is designed for high speed processing on tape lines and ideal material for low denier raffia tapes. It is formulated with additive package for low water carry over.

BIS Designation Code: IS 10951-1-TB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	4.5
Density	ASTM D792	g/ cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	32
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1350
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values. Not to be taken as specifications. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 210 – 250 °C

Water bath temperatures: 30 – 50 °C

Stretching temperatures: ~ 155 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Tapes for geotextiles, Packaging sacks and low weight general purpose wrapping fabrics.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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H050MN

HOMOPOLYMER

FOR THERMOFORMED AND INJECTION MOLDED PRODUCTS

Repol H050MN is PP homopolymer grade recommended for injection molding, sheet extrusion and thermoforming. It is used for the preparation of furniture compounds. It contains nucleating agent for high rigidity and reduced cycle time. It also contains antistatic agent that reduces static charge build up on products.

BIS Designation Code: IS 10951-1-MBL-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3.4
Density	ASTM D792	g/ cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	36
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1750
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	35
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	110
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 220 – 260 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Furniture and Rigid Packaging, Housewares and Thermoware.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H060FU

HOMOPOLYMER

FOR FILAMENTS, YARNS & NON-WOVENS

Repol H060FU is PP homopolymer grade recommended for Staple fibers and Continuous High Tenacity Filaments. The grade is specially formulated for Anti Gas fading and UV Stability. It is an ideal material for spinning of fine denier staple fibers.

BIS Designation Code: IS 10951-1-LBD-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	6
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1400
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 220 – 260 °C

Quenching Temperature: 10 – 25 °C

Draw Ratio: ~ 1.6

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Staple fibers for thermal bonded and needle punched nonwovens, High tenacity filament yarns.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H070SG

HOMOPOLYMER

FOR CAST FILM APPLICATION

Repol H070SG is PP homopolymer film grade specially developed for manufacturing of cast films. It exhibits very good transparency and stiffness when processed on cast film line.

BIS Designation Code: IS 10951-1-KB-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230 ° C/2.16 kg)	ASTM D1238	g/10 min	7.0
Density	ASTM D792	g/ cc.	0.90
FILM PROPERTIES (CAST FILM)			
Tensile Strength at Break (MD / TD)	ASTM D882	MPa	42 / 38
Elongation at Break (MD / TD)	ASTM D882	%	500 /550
Haze – 20 µ / 40 µ	ASTM D1003	%	0.7 / 1.3
Gloss @ 20 °	ASTM D2457	GU	90
Coefficient of friction (St / Kn)	ASTM D1894	---	> 0.7/ >0.7

* Typical values. Not to be taken as specifications

** Mechanical properties are on 40 microns. Film processed on cast film plant with chill roll temperature of 20 ° C.

Note: Final film properties are strongly influenced by the processing conditions.

Typical Processing Conditions

Extrusion temperatures: 210 – 250 ° C

Applications

Metallized films, Cast films for Food packaging, textile packaging, Lamination, stationary etc.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H080EY

HOMOPOLYMER

FOR TQ (TUBULAR WATER QUENCH) FILM

Repol H080EY is PP homopolymer grade recommended for TQ (Tubular Water Quench) Film. The grade is formulated with slip and antiblock additives.

BIS Designation Code: IS 10951-1-NBH-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
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Melt Flow Rate (230° C/2.16 kg)	ASTM D1238	g/10 min	8.0
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Density	ASTM D792	g/ cc.	0.90
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FILM PROPERTIES (TQPP FILM)

Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	23 / 20
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Tensile Strength at Break (MD / TD)	ASTM D882	MPa	34 / 33
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Elongation at Break (MD / TD)	ASTM D882	%	550 / 750
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Haze – 20 µ / 40 µ	ASTM D1003	%	3.2 / 8.0
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Gloss @ 20 ° / 60 °	ASTM D2457	GU	70 / 115
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Coefficient of friction (St / Kn)	ASTM D1894	---	0.25 / 0.20
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* Typical values. Not to be taken as specifications

Mechanical and optical properties are on 40 microns. Film processed on TQPP film plant with quench water temperature of 15 deg. C.

Typical Processing Conditions

Extrusion temperatures: 200 – 240 °C

Cooling water temperature: 15 – 25 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Textile overwrap films, Garment bags, Snack food, Grocery and General-purpose packaging.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H100EY

HOMOPOLYMER

FOR TQ (TUBULAR WATER QUENCH) FILM

Repol H100EY is PP homopolymer grade recommended for use in TQ (Tubular Water Quench) Film. The grade is specially formulated with slip and antiblock additives.

BIS Designation Code: IS 10951-1-NBH-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
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Melt Flow Rate (230 ° C/2.16 kg)	ASTM D1238	g/10 min	11.0
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Density	ASTM D792	g/ cc.	0.90
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FILM PROPERTIES (TQPP FILM)

Tensile Strength at Yield (MD / TD)	ASTM D882	MPa	22 / 19
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Tensile Strength at Break (MD / TD)	ASTM D882	MPa	32 / 30
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Elongation at Break (MD / TD)	ASTM D882	%	580 / 750
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Haze – 20 μ / 40 μ	ASTM D1003	%	3.0 / 7.0
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Gloss @ 20 ° / 60 °	ASTM D2457	GU	70 / 115
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Coefficient of friction (St / Kn)	ASTM D1894	---	0.25 / 0.20
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* Typical values. Not to be taken as specifications

Mechanical and optical properties are on 40 microns. Film processed on TQPP film plant with quench water temperature of 15 ° C.

Typical Processing Conditions

Extrusion temperatures: 200 – 240 ° C

Cooling water temperature: 15 – 25 ° C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Textile overwrap films, Garment bags, Snack food, Grocery and General-purpose packaging.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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H110FG

HOMOPOLYMER

FOR STAPLE FIBER AND FILAMENT

Repol H110FG is PP homopolymer grade recommended for making staple fiber and high tenacity filaments. It has excellent uniformity needed for high-speed spinning of fine denier fibers. The grade is specially formulated with the additive for Anti Gas fading.

BIS Designation Code: IS 10951-1-LB-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	8
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1600
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 220 – 260 °C

Quenching Temperature: 10 – 25 °C

Draw Ratio: 1.2 – 1.6

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Staple fibers for thermal bonded and needle punched nonwovens, High tenacity filament yarns.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H110FU

HOMOPOLYMER

FOR STAPLE FIBER AND FILAMENT

Repol H110FU is PP homopolymer grade recommended for making staple fiber and high tenacity filaments. It is suitable for high speed spinning of fine denier fibers. The grade is specially formulated for Anti Gas fading and UV stability.

BIS Designation Code: IS 10951-1-LBD-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	8
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1600
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 220 – 260 °C

Quenching Temperature: 10 – 25 °C

Draw Ratio: ~ 1.6

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Staple fibers for thermal bonded and needle punched nonwovens, High tenacity filament yarns.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

H110MA

HOMOPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol H110MA is PP homopolymer grade recommended for Injection Molding processes. It is suitable for making furniture compounds. The grade is formulated with antistatic agent that reduces static charge build-up in the products.

BIS Designation Code: IS 10951-1-MBJ-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	11
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	36
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1650
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	27
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 260 °C

Mold temperature: 30 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Rigid packaging, furniture & houseware

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H110MAS

HOMOPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol H110MAS is a nucleated PP homopolymer grade recommended for Injection Molding processes. It offers a unique combination of high crystallization temperature in the polymer and isotropic shrinkage in the final product. This blend enhances production rates across various applications.

In addition, H110MAS exhibits higher flexural modulus and surface gloss, facilitating the molding of high-quality parts by minimizing warpage, sink marks, and other molding defects.

BIS Designation Code: IS 10951-1-MBL-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	11
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	38
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1650
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	30
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	110
Hardness – Shore D	ASTM D2240	---	71

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 200 – 250 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Injection molded articles for rigid packaging, furniture & houseware.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H200FG

HOMOPOLYMER

FOR FIBER AND FILAMENT

Repol H200FG is PP homopolymer grade recommended for use in Fiber Spinning applications, for multifilament yarn and staple fiber. It is suitable for high speed spinning of fine denier fibers. It contains process stabilizer package suitable for non-woven and filament products. The grade is specially formulated with the additive for Anti Gas fading.

BIS Designation Code: IS 10951-1-LB-D

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	20
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	11
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1650
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	27
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 210 – 250 °C

Quenching Temperature: 10 – 20 °C

Godet rolls Temperature: 80 – 120 °C

Draw Ratio: ~ 3.2

Texturizing Temperature: 125 – 165 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Multifilament yarn, staple fiber & non-woven fabrics

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H200MA

HOMOPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol H200MA is PP homopolymer grade recommended for Injection Molding processes. The grade offers superior stiffness and formulated with antistatic agent that reduces static charge build-up in the products.

BIS Designation Code: IS 10951-1-MBJ-D

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	25
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	32
Elongation at Yield (50 mm/min.)	ASTM D638	%	11
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1650
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	27
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 250 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Thin walled containers, Multicavity molding and Articles with intricate long flow paths.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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H200MK

HOMOPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol H200MK is PP homopolymer grade recommended for Injection Molding processes. It offers good stiffness. The grade is specially formulated with clarifier and antistatic agent that reduces static charge build-up in the products.

BIS Designation Code: IS 10951-1-MBL-D

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	20
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	11
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1700
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	26
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	120
Hardness – Shore D	ASTM D2240	---	69

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 200 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Syringes, multicavity molding and Articles with intricate long flow paths.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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H350EC

HOMOPOLYMER

FOR EXTRUSION COATING / LAMINATION

Repol H350EC is a PP Homopolymer grade recommended for use in Extrusion Coating / Lamination process. The grade is designed to provide excellent flow characteristics, low neck-in & consistent processability, good surface finish, gloss and excellent adhesion to variety of substrates.

BIS Designation Code: IS 10951-1-HB-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	38
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	9
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1500
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	24
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 220 – 260 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Recommended to use for below mentioned applications meeting ash content limit as per Bureau of Indian Standards (BIS) / any other relevant standards. Recommend to use optimum percentage of extrusion coating grade of Low-density polyethylene (LDPE) for required GSM / thickness and line speed.

Applications

Coating of Polypropylene Woven Fabrics, Compounding and Masterbatch.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H350FG

HOMOPOLYMER

FOR FIBER AND FILAMENTS/ LAMINATION/ NON-WOVEN

Repol H350FG is PP homopolymer grade recommended for use in fiber and filaments, spun bond non-woven, extrusion coating. It is suitable for high speed spinning of fine denier fibers and low GSM non-woven fabrics. The grade is specially formulated for Anti Gas fading properties.

BIS Designation Code: IS 10951-1-CB-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	38
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	9
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1500
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	24
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 180 – 260 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Can be used for extrusion coating/ lamination on plastic woven sacks meeting ash content limit as per Bureau of Indian Standards (BIS) / any other relevant standards. Recommend to use optimum percentage of extrusion coating grade of Low-density polyethylene (LDPE) for required GSM / thickness and line speed.

Applications

POY, FDY & Crimped filament yarns, Spun-bond nonwoven fabrics for Filtration, Hygiene & Medical Disposables, Packaging and Agriculture.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

DISCLAIMER

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H350FH

HOMOPOLYMER

FOR FIBER & FILAMENT AND NON-WOVEN

Repol H350FH is PP homopolymer grade recommended for use in fiber & filaments and spun bond non-woven. It is suitable for high speed spinning of fine denier fibers and low GSM non-woven fabrics. The grade is specially formulated for Anti Gas Fading properties.

BIS Designation Code: IS 10951-1-CB-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	38
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	34
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1550
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	22
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	104
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion Temperature: 180 – 260 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

POY, FDY & Crimped filament yarns, Spun-bond nonwoven fabrics for Filtration, Hygiene & Medical Disposables, Packaging and Agriculture.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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SS35N

HOMOPOLYMER

FOR WOVEN SACKS & MONOFILAMENTS

Repol SS35N is a PP homopolymer grade recommended for making raffia tapes used in making plastic woven sacks. It exhibits good processability with excellent mechanical properties. The grade is formulated with additive package for low water carry over.

BIS Designation Code: IS 10951-1-TB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	35
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1400
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	45
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	70

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Extrusion temperatures: 220 – 260 °C

Water bath temperatures: 30 – 50 °C

Stretching temperatures: ~ 155 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Woven fabrics, Sacks & Monofilaments, Carpet backing, Industrial packaging and Geotextiles.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA:CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight

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B030MG

IMPACT COPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol B030MG is PP impact copolymer grade recommended for use in Injection molding where medium impact strength is desired. It exhibits excellent processability with impact - stiffness balance.

BIS Designation Code: IS 10951-3-MB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3.3
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	27
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1000
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	150
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Furniture, Crates, Automotive components & Industrial products.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

B120MA

IMPACT COPOLYMER

FOR INJECTION MOULDED PRODUCTS

Repol B120MA is PP impact copolymer grade recommended for use in Injection Molding where medium impact strength is required. It exhibits good processability and good impact – stiffness balance. The grade is specially formulated with antistatic agent.

BIS Designation Code: IS 10951-3-MBJ-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	24
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1000
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	85
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Furniture, Industrial Components, General Purpose Molding.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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B220MN

IMPACT COPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol B220MN is PP impact copolymer grade recommended for use in Injection Molding where high flow and medium impact strength are desired. It exhibits good processability with high flow and good impact – stiffness balance. The grade is specially formulated with nucleating agent.

BIS Designation Code: IS 10951-3-MBJ-D

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	22
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	25
Elongation at Yield (50 mm/min.)	ASTM D638	%	8
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1200
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	80
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Rigid Packaging, Automotive Components, Housewares and Appliances Parts.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with the FDA : CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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B400MN

IMPACT COPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol B400MN is PP impact copolymer grade recommended for use in Injection Molding where high flow and medium impact strength are desired. The grade is specially formulated with nucleating agent.

BIS Designation Code: IS 10951-3-MBJ-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	38
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	26
Elongation at Yield (50 mm/min.)	ASTM D638	%	7
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1150
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	70
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Appliances, Automotive, Compounding & Thin Wall Injection Molded Products.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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B650MN

IMPACT COPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol B650MN is PP impact copolymer grade recommended for Injection Molding where high flow and medium impact strength are desired. The grade is specially formulated with nucleating agent.

BIS Designation Code: IS 10951-3-MB-F

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	70
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	24
Elongation at Yield (50 mm/min.)	ASTM D638	%	4
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1300
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	70
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Appliances, Automotive, Compounding & Thin Wall Injection Molded Products.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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Reliance Industries Limited

<i>P O L Y P R O P Y L E N E</i>		
Provisional Datasheet		
REPOL B300MN		Impact Copolymer

Product description & applications :

REPOL Polypropylene B300MN is manufactured using Unipol PP process which combines the production efficiency of gas phase fluidized bed reactor technology with the high activity stereospecific catalyst system.

Repol B300MN is high crystalline Impact Copolymer grade recommended for use in **Injection Moulding processes**. The grade contains nucleating agent. It is an ideal material for making automobile parts where high flow with very high stiffness is required. The grade is also suitable as a compounding base for automotive and appliance sectors.

Features :

Repol B300MN is designed to provide :

- High Stiffness , medium Impact
- Fast Moulding cycles
- Excellent flow characteristics
- Good processing stability
- High crystallinity

Resin Properties	ASTM		
	Method	Typical Values*	
		Unit	Value
Melt Flow Rate (230°C/2.16 kg)	D1238	g /10min.	30
Tensile Strength @ Yield (50mm/min)	D638	MPa	30
Elongation @ Yield (50mm/min)	D638	%	5
Flexural Modulus (ASTM :1% secant, 1.3mm/min)	D790A	MPa	1650
Izod Impact Strength Notched, @23°C	D256	J/m	60
Charpy Impact @ 23°C	D256	J/m	70
Heat Deflection Temperature (@ 455KPa)	D648	°C	115

Repol B300MN meets the requirements stipulated in IS 10910 on “Specification for Polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals, and drinking water”. Additives incorporated in this grade conform to the positive list of constituents as prescribed in IS-10909. The grade and the additives incorporated in it also comply with the FDA:CFR Title 21,177.1520, Olefin polymers.

REPOL is the registered trademark for Polypropylene from Reliance Industries Limited

** Typical values with injection moulded specimens, not to be taken as specification*

The data, information and suggestions contained herein are given purely as a guide. Reliance Industries Limited and Reliance Group Companies undertake no responsibility either for the results deriving from their adoption or for possible positions in apparent contrast with existing patent rights

C015EG

IMPACT COPOLYMER

FOR EXTRUSION & BLOW MOLDED PRODUCTS

Repol C015EG is PP impact copolymer grade recommended for use in Blow molding and extrusion where high impact strength is desired. It exhibits excellent processability and rigidity.

BIS Designation Code: IS 10951-3-EB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	1.8
Density	ASTM D792	g/cc.	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	27
Elongation at Yield (50 mm/min.)	ASTM D638	%	11
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1000
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	200
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Thermoforming temperature: 130 – 150 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Sheets, Blow molded containers, Thermoformed Articles, Corrugated board, Automotive Trims etc.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

C080MA

IMPACT COPOLYMER

FOR INJECTION MOULED PRODUCTS

Repol C080MA is PP impact copolymer grade recommended for use in Injection Molding where high impact strength is desired. It exhibits good processability and impact – stiffness balance. The grade is specially formulated with antistatic agent.

BIS Designation Code: IS 10951-3-MBJ-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	9
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	23
Elongation at Yield (50 mm/min.)	ASTM D638	%	7
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1000
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	150
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Furniture, Luggage, Paint Pails, Industrial components.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

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C080MT

IMPACT COPOLYMER

FOR INJECTION MOULED PRODUCTS

Repol C080MT is PP impact copolymer grade recommended for use in Injection Molding where high impact strength is desired. It exhibits good processability and impact – stiffness balance. The grade is specially formulated with long term heat stabilizer required for under the hood applications in automotive.

BIS Designation Code: IS 10951-3-JBC-B

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	9.5
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	24
Elongation at Yield (50 mm/min.)	ASTM D638	%	7
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1050
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	100
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	100
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Battery containers, Automotive, Industrial components.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

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C120MN

IMPACT COPOLYMER

FOR INJECTION MOULED PRODUCTS

Repol C120MN is PP impact copolymer grade recommended for use in Injection Molding where good impact stiffness balance is desired. The grade is specially formulated with antistatic agent.

BIS Designation Code: IS 10951-3-MBJ-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	24
Elongation at Yield (50 mm/min.)	ASTM D638	%	7
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1050
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	150
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	105
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Paint pails, Luggage, Industrial components.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

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C320MN

IMPACT COPOLYMER

FOR INJECTION MOULED PRODUCTS

Repol C320MN is PP impact copolymer grade recommended for Injection Molding where high flow and medium impact strength is desired. The grade is specially formulated with nucleating agent.

BIS Designation Code: IS 10951-3-MBJ-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	32
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	20
Elongation at Yield (50 mm/min.)	ASTM D638	%	5
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1150
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	150
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	92
Hardness – Shore D	ASTM D2240	---	68

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Appliances, Automotive and Compounding.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

D120MA

IMPACT COPOLYMER

FOR INJECTION MOULED PRODUCTS

Repol D120MA is PP impact copolymer grade recommended for Injection Molding where extra high impact strength is desired. It exhibits no break during impact and good stiffness. The grade is specially formulated with antistatic agent.

BIS Designation Code: IS 10951-3-MBJ-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	19
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	900
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	No Break
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	90
Hardness – Shore D	ASTM D2240	---	66

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Automotive components, Industrial products, Luggage shells.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.



D350MN

IMPACT COPOLYMER

FOR INJECTION MOULDED PRODUCTS &
COMPOUNDING

Repol D350MN is PP impact copolymer grade recommended for Injection Molding & Compounding. It exhibits high flow with no break during impact. The grade is specially formulated with antistatic agent and nucleating agent.

BIS Designation Code: IS 10951-3-MBJ-E

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	35
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	19
Elongation at Yield (50 mm/min.)	ASTM D638	%	10
Flexural Modulus (1% Secant)	ASTM D790A	MPa	990
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	No Break
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	90
Hardness – Shore D	ASTM D2240	---	66

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Automotive components, Appliance components, Base resin for Compounding, Industrial products.

Regulatory Information

D350MN is not recommended for Food contact and medical applications.
Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

MI3530

IMPACT COPOLYMER

FOR INJECTION MOLDED PRODUCTS

Repol MI3530 is PP impact copolymer grade recommended for Injection Molding where medium impact strength is required. It exhibits excellent processability with impact - stiffness balance.

BIS Designation Code: IS 10951-3-MB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	3
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	26
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1100
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	115
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	69

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 240 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Furniture, Crates, Textiles, Automotive components & Industrial products.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.

AER003N

RANDOM COPOLYMER FOR EXTRUDED PIPES & SHEETS

Repol AER003N is PP random copolymer recommended for pipe extrusion, sheet for thermoforming and extrusion blow molding.

BIS Designation Code: IS 10951-2-PB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	0.3
Melt Flow Rate (230°C/5.0 kg)	ASTM D1238	g/10 min	1.0
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	27
Elongation at Yield (50 mm/min.)	ASTM D638	%	14
Elongation at Break (50 mm/min.)	ASTM D638	%	> 500
Flexural Modulus (1% Secant)	ASTM D790A	MPa	850
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	No Break
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	92

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Pipe extrusion

Melt temperature: 190 – 220 °C

Water bath temperature: 20 – 40 °C

Injection molding of fittings

Melt temperature: 190 – 240 °C

Mold temperature: 30 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Pipes, Fittings, Sheets & Thermoforming

Note:

- Repol AER003N is a natural PP resin for PPR pressure Pipes application. The material is lightly stabilized and meets the designation of PP-R-P-A-N-10-35-003 as per IS 10951.
- When compounded with an approved and adequate quantity of antioxidants / thermal stabilizers it meets the designation of PP-R-P-H-N-10-25-003 as per IS 10951 and also meets the performance criteria as per IS 15801.
- When compounded with required colorants along-with approved and adequate quantity of antioxidants / thermal stabilizer It meets the designation of PP-R-P-H-C-10-25-003 as per IS 10951 and also meets the performance criteria as per IS 15801.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

The information contained herein may include typical properties and processing parameters of the grade or its typical performances when used in respective applications. The values given above are based on analysis of representative samples and not the actual product supplied. It is the customer's responsibility to inspect and test our grades in order to satisfy itself as to the suitability of the products for customers' particular application. The customer is solely responsible for all determinations regarding any use of material or product and any process in its area of interest. RIL assumes no obligation or liability for any loss, damage or injury directly or indirectly suffered or incurred as a result of using any of the information or product given in this document. The information and data presented herein is true and accurate to the best of our knowledge. No warranty or guarantee expressed or implied, is made regarding performance or otherwise. This information and data may not be considered as a suggestion to use our products without taking into account existing patents, or legal provisions or regulations, whether national or international. The user of any information and/or data is advised to obtain the latest details from any of the offices of the company or its authorized agents, as the information and/or data is subject to change based on the research and development work undertaken by the company.



SR20NC

RANDOM COPOLYMER

FOR BLOW MOULDING & EXTRUSION

Repol SR20NC is PP random copolymer grade recommended for Blow Molding and Extrusion. It provides very good processability with excellent melt strength. The grade is specially formulated with clarifier.

BIS Designation Code: IS 10951-2-BBL-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	1.9
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	30
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	900
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	100
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 180 – 240 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

High clarity bottles, containers, sheets, ball pen refills, stationery items and thermoformed products

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with the FDA : CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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SR20NS

RANDOM COPOLYMER

FOR BLOW MOLDING & EXTRUSION

Repol SR20NS is PP random copolymer grade recommended for Blow Molding and Extrusion. It provides very good processability with excellent melt strength. The grade combines good clarity and gloss.

BIS Designation Code: IS 10951-2-BB-A

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	1.9
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	26
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	800
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	65
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	90
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 180 – 240 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Blow molded containers for cosmetic and personal care products, extrusion.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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SRM100NC

RANDOM COPOLYMER

FOR INJECTION MOLDING &
STRETCH BLOW MOLDING

Repol SRM100NC is PP Copolymer recommended for Injection Molding and stretch blow molding process. The grade is specially formulated with clarifier and antistatic agent.

BIS Designation Code: IS 10951-2-MBL-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	30
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1100
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	65
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 260 °C

Mold temperature: 30 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Rigid containers, Houseware, ISBM bottles, syringes.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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SRM250NC

RANDOM COPOLYMER

FOR INJECTION MOLDING &
STRETCH BLOW MOLDING

Repol SRM250NC is PP random copolymer grade recommended for Injection Molding and Injection stretch blow molding. It exhibits high clarity and gloss. The grade is specially formulated with clarifier and antistatic agent.

BIS Designation Code: IS 10951-2-MBL-D

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	30
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	29
Elongation at Yield (50 mm/min.)	ASTM D638	%	9
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1000
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	50
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	100
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 190 – 260 °C

Mold temperature: 20 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

TWIM, Rigid containers, High Clarity Containers, Houseware, ISBM bottles, Syringes.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers.

Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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SRX100

RANDOM COPOLYMER

ULTRA HIGH CLARITY GRADE FOR
INJECTION & STRETCH BLOW MOULDING

Repol SRX100 is PP random copolymer grade recommended for Injection Molding and Injection Stretch Blow Molding. It exhibits excellent clarity and processability. The grade is specially formulated with clarifier and antistatic agent.

BIS Designation Code: IS 10951-2-MBL-C

Typical Characteristics

Property	Test Method	Unit	Typical Value*
Melt Flow Rate (230°C/2.16 kg)	ASTM D1238	g/10 min	12
Density	ASTM D792	g/cc	0.90
Tensile Strength at Yield (50 mm/min.)	ASTM D638	MPa	30
Elongation at Yield (50 mm/min.)	ASTM D638	%	12
Flexural Modulus (1% Secant)	ASTM D790A	MPa	1100
Notched Izod Impact Strength (23 °C.)	ASTM D256	J/m	65
Heat Deflection Temperature (455 KPa)	ASTM D648	°C	95
Hardness – Shore D	ASTM D2240	---	67

* Typical values, not to be taken as specification. All the mechanical properties as per ASTM D638 injection molded Type 1 specimen in accordance with ASTM D4101

Typical Processing Conditions

Melt temperature: 210 – 260 °C

Mold temperature: 30 – 50 °C

Note: Processing parameters mentioned above are for reference only and not to be considered as specifications. They may vary based on the product to be manufactured.

Applications

Ultra-high clarity rigid containers, houseware, ISBM bottles, appliance parts.

Regulatory Information

The product complies with Indian Standard IS 10910 on “Specification for polypropylene and its copolymers for safe use in contact with foodstuffs, pharmaceuticals and drinking water. It also conforms to IS 16738:2018 on positive list of constituents for polypropylene, polyethylene and their copolymers for its safe use in contact with foodstuffs and pharmaceuticals. The grade and the additives incorporated in it also comply with FDA : CFR Title 21,177.1520, Olefin polymers. Not to be used in the manufacture of Single Use Plastic (SUP) items prohibited under PWM Rules, 2016.

Storage Recommendations

Bags should be stored in dry / closed conditions at temperatures below 50°C and protected from UV / direct sunlight.

DISCLAIMER

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